Quantum Computing
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Lecture Topic: TOPIC

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In this lecture, we recall basic definitions and concepts of probability theory.

1 Events, Probability, Probability Space

Consider an experiment with a finite (or countably infinite) number of outcomes. Each outcome is a simple event (or a sample point). The *sample space* is the set of all possible simple events. An *event* $\mathcal{E}$ is a union of simple events — a subset of the sample space.

Two events are said to be **mutually exclusive** if $A \cap B = 0$.

With each simple event s we associate a number $\mathbb{P}(s)$ which is called as the *probability* of s.

Definition 1 ▶ Probability Space

A probability distribution $\mathbb{P}$ on a discrete sample space S is a function from events of S to $\mathbb{R}$ such that it satisfies the following three *axioms*:

1. $\mathbb{P}(A) \geqslant 0$ for any event A.
2. $\mathbb{P}(S) = 1$.
3. For any (finite or countably infinite) sequence of pairwise mutually exclusive events $A_1, A_2, \ldots$:
$$\mathbb{P}(\cup_i A_i) = \sum_i \mathbb{P}(A_i)$$

The pair $(S, \mathbb{P})$ is called a discrete **probability space**.

The probability of any event $\mathcal{E}$ can be computed as the sum of the probabilities of the simple events that it is composed of; this follows from the third axiom:

$$\mathbb{P}(\mathcal{E}) = \sum_{s \in \mathcal{E}} \mathbb{P}(s)$$

2 Principle of Inclusion-Exclusion

The *inclusion-exclusion* principle gives a formula for computing the probability of the union of a set of events in terms of the probabilities of the individual events and their intersections.

Theorem 1 ▶ Inclusion-Exclusion Principle

Let $E_1, E_2, \ldots, E_n$ be arbitrary events. Then

$$\begin{aligned}\mathbb{P}(\cup_{i=1}^{n} E_i) &= \sum_{i=1}^{n} \mathbb{P}(E_i) - \sum_{1 \leqslant i < j \leqslant n} \mathbb{P}(E_i \cap E_j) + \sum_{1 \leqslant i < j < k \leqslant n} \mathbb{P}(E_i \cap E_j \cap E_k) \\ &\quad - \cdots + (-1)^n \mathbb{P}(E_1 \cap E_2 \cdots \cap E_b) \\ &= \sum_{k=1}^{n} (-1)^{k+1} \left(\sum_{1 \leqslant i_1 < \cdots < i_k \leqslant n} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \cdots \cap E_{i_k}) \right)\end{aligned}$$

An important consequence of the inclusion-exclusion principle is a simple upper bound on the probability of the union of events known as *Boole's inequality or union bound*.

Theorem 2 ▶ Boole's inequality (union bound)

For any arbitrary sequence of events $E_1, E_2, \ldots, E_n$:

$$\mathbb{P}(\cup_{i=1}^{n} E_i) \leqslant \sum_{i} \mathbb{P}(E_i)$$

The union bound is used often in upper bounding the occurrence of the union of a set of "bad" events. If this upper bound is small, then it implies that none of the bad events occur with high probability.

3 Conditional Probability

Definition 2 ▶ Conditional Probability

Given two events A and B, the conditional probability of A given B is defined as follows:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

assuming $\mathbb{P}(B) \neq 0$.

In the above definition, by conditioning on B we restrict the sample space to the set B. Thus the conditional probability can be considered as $\mathbb{P}(A \cap B)$ "normalized" by $\mathbb{P}(B)$.

Given two events E_1 and E_2, Bayes' rule relates the conditional probability of the first given the second, to the conditional probability of the second given the first. This is useful to infer one conditional probability from the other.

Theorem 3 ▶ Bayes' rule

$$\mathbb{P}(E_1 \mid E_2) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_2)} = \frac{\mathbb{P}(E_2 \mid E_1)\,\mathbb{P}(E_1)}{\mathbb{P}(E_2)}$$

The following are some useful identities that involve computing the probability of the intersection of many events.

When you compile this theorem looks ugly. Not clear why?

Theorem 4 ▶ Chain rule of Conditional Probabilities

- $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B)$.
- $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A \mid B \cap C)\,\mathbb{P}(B \cap C) = \mathbb{P}(A \mid B \cap C)\,\mathbb{P}(B \mid C)\,\mathbb{P}(C)$.
- The following is a generalization of the above identity. Let $A_1, \ldots, A_n$ be a sequence of events. Let $E_i = \bigcap_{j=1}^{i} A_i$. Then

$$\begin{aligned}\mathbb{P}(E_n) &= \mathbb{P}(A_n \mid E_{n-1})\,\mathbb{P}(E_{n-1}) \\ &= \mathbb{P}(A_n \mid E_{n-1})\,\mathbb{P}(A_{n-1} \mid E_{n-2}) \ldots \mathbb{P}(A_2 \mid E_1)\,\mathbb{P}(A_1)\end{aligned}$$

A fundamental concept that follows from conditional probability is that of *independence*.

Definition 3 ▶ Independence of events

Two events A and B are said to be *independent* if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \times \mathbb{P}(B)$$

or (when $\mathbb{P}(B) > 0$)

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} = \mathbb{P}(A)$$

4 The Birthday Paradox

We study a problem called the "Birthday Paradox", that arises often in algorithm design and analysis. The problem also serves to illustrate basic concepts in probability theory.

Question: What is the probability that among m people no two have the same birthday? We make the following two assumptions: (1) All birthdays are equally likely and (2) Birthdays are independent events.

To compute the above probability, we first define the sample space of the experiment. The sample space is the set of all vectors $S = \{(b_1, \ldots, b_m) | b_i \in [1, \ldots, N]\}$, where b_i denotes the birthday of the ith person, and N is the total number of different birthdays ($N = 365$ in Earth). We need to compute $\mathbb{P}(E)$ where the event $E = \{(b_1, \ldots, b_m | b_i \neq b_j \text{ for all } i \neq j\}$, i.e., the event E is the set of all events where no two birthdays are the same.

How many different atomic events are counted in E? The number of possible m different birthdays is $N.(N-1).(N-2)\ldots(N-m+1)$. Hence,

$$\begin{aligned}\mathbb{P}(E) &= \frac{N \cdot (N-1) \cdot (N-2) \ldots (N-m+1)}{N^m} \\ &= \prod_{i=0}^{m-1} (1 - i/N) \\ &\leqslant \prod_{i=0}^{m-1} e^{-i/N} \\ &= e^{-\sum_{i=0}^{m-1} i/N} \\ &= e^{-m(m-1)/2N}\end{aligned}$$

For $m = \sqrt{2N} + 1 \leqslant 28$, $\mathbb{P}(E) < 1/e < 1/2$.

The apparent "paradox" in this problem is that significantly less number of people, i.e., only about $\sqrt{N}$ people (which is much smaller than N), are needed to have a good chance (about 50%) to have two people to have a common birthday.

Alternate Analysis

Assume that we choose one birthday after the other independently and uniformly at random from $[1, 2, \ldots, N]$. Let the event E_i denote: "the ith choice is different from the first $i-1$ choices". Then, we compute:

$$\begin{aligned}
\mathbb{P}(E) &= \mathbb{P}(\cap_{i=1}^{m} E_i) \\
&= \mathbb{P}(E_1)\,\mathbb{P}(E_2 \mid E_1)\,\mathbb{P}(E_3 \mid E_2 \cap E_1) \ldots \mathbb{P}\Big(E_m \mid \cap_{i=1}^{m-1} E_i\Big) \\
&= 1(1-1/N)(1-2/N)\ldots(1-(m-1)/N) \\
&= \prod_{i=1}^{m}(1-\frac{i-1}{N})
\end{aligned}$$

This gives the same result as before.

This analysis uses a very useful principle called the **Principle of Deferred Decisions**. In this principle, the idea is to defer the fixing of choices of events to "when they are needed" and not a priori (i.e., all at once). Typically, this type of analysis will use conditional probabilities. In the above analysis, to compute $\mathbb{P}(E)$, we first computed $\mathbb{P}(E_1)$, then $\mathbb{P}(E_2 \mid E_1)$ and so on. At each stage, we only fixed the event that determined the respective conditional probabilities, i.e., first E_1 which fixes b_1, and then the event E_2 conditioned on E_1, which fixes $b_2 \neq b_1$, and so on.

5 Latex resources

An easy way to write latex is using overleaf: http://overleaf.com.

See the latex tutorials at https://www.overleaf.com/learn/latex/Tutorials

You can also find various latex manuals and documentation at https://latex-project.org/guides/.

6 Theorems

Theorems that we discuss in class should be stated correctly and proved. Here is an example.

Markov's inequality uses only the value of the expectation (also called "first moment"). Hence it can be quite weak.

Theorem 5 ▶ Markov Inequality

For any non-negative random variable and for any $a > 0$

$$\mathbb{P}(X \geqslant a) \leqslant \frac{\mathbb{E}[X]}{a}$$

Except for the easiest theorems and proofs, it is important to give an intuition for them. Here is an example.

Intuitively, Markov's inequality says that for any (positive) random variable X, it is less likely that X takes a value much larger than its expectation.

Proof. For any $a > 0$, let I be an indicator r.v. for the event $X \geqslant a$. Then $I \leqslant X/a$. Taking expectations on both sides, we get $\mathbb{E}[I] = \mathbb{P}(X \geqslant a) \leqslant \frac{\mathbb{E}[X]}{a}$. □

Chebyshev's inequality gives a stronger bound, which assumes that the *variance* (also called the "second moment") or *standard deviation* of the random variable is known. These are defined below.

It is good to define essential concepts formally by defining them. Here is an example.

Definition 4

The **variance** of a random variable X is

$$\mathbb{V}[X] = \mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]$$

The **standard deviation** of a random variable X is

$$\sigma(X) = \sqrt{\mathbb{V}[X]}$$

Now we state and prove Chebyshev's inequality.

Theorem 6

Chebyshev's Inequality For **any** random variable

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) \leqslant \frac{\mathbb{V}[X]}{a^2}$$

The proof uses Markov's inequality. Note that Chebyshev's inequality applies to any random variable, not necessarily positive. Thus to apply Markov (which applies only to positive r.v.) we apply it to $|X - \mathbb{E}[X]|$. The usual trick to get rid of absolute value, which is hard to deal with, is by squaring.

Proof.

$$\mathbb{P}(|X - \mathbb{E}[X]| \geqslant a) = \mathbb{P}\Big((X - \mathbb{E}[X])^2 \geqslant a^2\Big)$$

By Markov inequality,

$$\mathbb{P}\Big((X - \mathbb{E}[X])^2 \geqslant a^2\Big) \leqslant \frac{\mathbb{E}\Big[(X - \mathbb{E}[X])^2\Big]}{a^2} = \frac{\mathbb{V}[X]}{a^2}$$

□

7 A Lemma in Latex

Lemmas are statements that help in proving a theorem (usually the main result). Lemmas also should be stated correctly and proved. This Lemma is from one of my papers.

Lemma 7

Let $C > 4D$, then at any given step $t \geqslant c \log N$, for a fixed constant $c > 0$, w.h.p. there is at least $\frac{1}{4}c \log N$ d-nodes in the network. Further, the above set of d-nodes arrived during the interval $[t - c \log N, t]$.

Proof. Let v be a node of degree $d(v)$ at time τ, then the expected number of neighbors of v that leave the network at time τ is $\frac{d(v)}{N}$. Thus, the expected number of new connections generated by v at this step is $\frac{d(v)}{N}\frac{D}{d(v)} = \frac{D}{N}$.

Consider a time interval $[t - c \log N, t]$. W.h.p. the network has no more than $N(1 + o(1))$ nodes at each step of this interval. Thus, the expected number of connections to the cache due to deletions during the whole interval is $D(1 + o(1))c \log N$. Each node that leaves the network contributes up to C connections to this count. We partition the count to C sums, each is a sum of independent $0 - 1$ events, and apply the Chernoff bound to each of the sums to

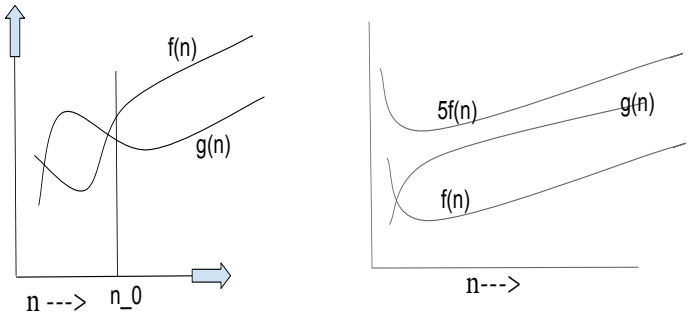


Figure 1: Illustration of the Big-O notation. $g(n) = O(f(n))$.

prove that the w.h.p the total number of connection to cache nodes from old nodes is bounded by $D(1 + o(1))c \log N$. The new arrivals during this interval have Poisson distribution with expectation $c \log N$, and each contribute D connections to cache node. Thus, w.h.p we have $c \log N(1 \pm o(1))$ new nodes and up to $D(1 + o(1))c \log N$ connections from these nodes. A cache node enters the cache with D connections and leaves with C connections, thus, w.h.p. this interval generates up to $c \log N \frac{2D(1+o(1))}{C-D} \leqslant \frac{2}{3} c \log N$ new c-nodes, and at least $\frac{1}{3} c \log N$ d-nodes. Since w.h.p. at least $c \log N(1 - o(1))$ d-node arrived in the interval, we have a total of $\frac{1}{4} c \log N$ in the network at time t. □

8 Figures and Pseudocode

8.1 Figures in Latex

You can write figures in latex as in Figure 1. One way of creating figure is to draw using some software (e.g., MS Paint, Photoshop) and save it as a vector graphic (e.g. svg, pdf). Then insert the file into the document using the figure environment. Here, Big-O.pdf and Big-O1.pdf are the figure files.

Alternatively, there are packages to draw figures directly. Tikz is a popular one, though it has a high learning curve. For an example, see here.

8.2 Algorithms in Latex

Pseudocode can be written in a few different ways. Using the algpseudocodex package, Algorithm Select is given in Algorithm 1.

9 Citations

Bibliographic information can be cited using `\cite`. For example, `\cite{algopandu}` cites the algorithm textbook [1]. Typically, we keep bibliographic information in a `.bib` file (in this case, `references.bib`).

Internal references (with hyperlinks) can be created using `\ref`, `\nameref`, and `\cref` (though `\cref` comes from the Cleveref package). The `\ref` command will cite the corresponding numerical label (e.g., `\ref{alg:select}` outputs "1"). The `\nameref` command will cite the name (if given), and `\cref` will output the numerical label with the type, e.g., `\cref{alg:select}` outputs "Algorithm 1".

Algorithm 1 Select

```
1: func SELECT(S, k):
2:     if |S| == 1:
3:         return S[1]
4:     Partition S into ⌈n/5⌉ groups of 5 elements each and a leftover group of up to 4 elements.
5:     Find the median of each group
6:     R = the set of these ⌈n/5⌉ medians
7:     m = |R|
8:     p = SELECT(R, ⌈m/2⌉)
9:     S_1 = {x ∈ S | x < p}
10:    S_2 = {x ∈ S | x > p}
11:    if |S_1| = k − 1:
12:        return p
13:    else if |S_1| > k − 1:
14:        return SELECT(S_1, k)
15:    else:
16:        return SELECT(S_2, k − |S_1| − 1)
```

References

[1] G. Pandurangan. *Algorithms: An Intuitive Approach.* Amazon, 2025.